

LESSON 10.10

USING TABLES TO DETERMINE IF A FUNCTION IS LINEAR



LESSON INTRODUCTION

MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

LEARNING TARGETS

- I can determine whether a function could represent a linear function given a table of values.
- I can determine whether a function is a linear function given an equation or graph.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input/output table, determine a function type that could represent it.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.

ESSENTIAL QUESTIONS

- How do you determine whether a function could represent a linear function given a table of values?
- How do you determine whether a function is a linear function given an equation or graph?

LESSON NARRATIVE

This lesson continues the focus on linear functions by analyzing input-output tables. Students determine if the rates of change are constant from tables of values and determine what the rate of change is for those that are constant. The students then verify that given graphs of the equations represented in the tables are linear functions. Then students analyze tables that have a constant rate of change but whose graphs do not represent linear functions. Conclusions are drawn that tables only give a snapshot of the entire function and therefore can only be used to determine if a linear function could be represented.

COMPONENT SUMMARY

10.10.1 WARM-UP (~5 MIN.)

Reason which one doesn't belong among four patterns of numbers

10.10.3 GUIDED INSTRUCTION (15-20 MIN.)

Recognize that tables can be used to conclude that a function could be linear

10.10.5 WRAP-UP (~5 MIN.)

Identify and explain which tables could be linear functions and which equation and graph are linear functions

10.10.2 GUIDED PRACTICE (15-20 MIN.)

Determine if rates of change are constant from tables of values and if graphs of equations are linear

10.10.4 YOUR TURN! (5 MIN.)

Determine if relations represented in tables could be linear functions and if given equations and graphs are linear functions

RESOURCES

GLOSSARY

Key Terms:

- Linear function

Additional Terms: N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- (Optional) Consider allowing students to use a graphing utility to graph the functions.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may calculate the rate of change as the horizontal change (in x) divided by the vertical change (in y).
- Students may make computational errors when calculating the rate of change.
- Students may confuse the constant rate of change with the constant term of an equation.
- Students may ignore the portion of the function not included in the table or on the graph and falsely assume the function is linear.

THINGS TO CONSIDER

- Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in determining whether a function could represent a linear function given a table of values, equation, or graph.
- If a student's computational skills are preventing them from accomplishing the goal of the lesson, consider allowing the student to use a calculator. To reinforce their computational skills, suggest to the student that they write and say each number and operation to themselves as they enter it in the calculator.
- Consider having students use Desmos to graph the functions. Desmos allows students to explore the domain and range outside of the values in the table.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

The routine Think-Pair-Share can be used in each component of the lesson. Initially, have students attempt the questions individually. Next, pair students up and ask them to discuss their strategies, reasoning, and answers. Tell students to resolve any differences in their answers. This routine helps students build their confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

The table of values, graphs, and equations provide students with

multiple representations of the function. During the lesson, students make connections between the representations and draw conclusions about how to determine whether they could represent a linear function.

DIFFERENTIATE INSTRUCTION

Consider having students create a reference card with an explanation of how to determine if a table of values, equation, or graph is a linear function. Encourage students to include at least one example of each process. Allow students to use their reference cards during the lesson.

The graphs in 10.10.2 Guided Practice to 10.10.5 Wrap-Up provide a visual entry point. Structuring components with the Think-Pair-Share routine provides an auditory entry point. Have students trace the graph with their fingers for a kinesthetic entry point.

Consider providing vocabulary terms that students must use in each response in the 10.10.5 Wrap-Up.

- Relation 1: Rate of Change
- Relation 2: Constant
- Relation 3: Horizontal
- Relation 4: Vertical

10.10.1 WARM-UP: WHICH ONE DOESN'T BELONG?

DIFFERENTIATE INSTRUCTION

Consider making this a Four Corners activity for kinesthetic engagement. Students should then find a partner from another corner to find for question 2.



10.10.1 Warm-Up: Which One Doesn't Belong?

Four number patterns are shown.

Pattern A	Pattern B
5, 9, 13, 17, 21, ...	5, 15, 45, 135, 405, ...
Pattern C	Pattern D
5, 2, -1, -4, -7, ...	10, 13, 16, 19, 22, ...

1. Complete the statement to explain which pattern doesn't belong.

- ☐ Pattern A
- ☐ Pattern B
- ☐ Pattern C
- ☐ Pattern D

doesn't belong because _____

Answers will vary.

- *Pattern A- The values are increasing by 4 instead of adding, subtracting or multiplying by 3.*
- *Pattern B- The values come from multiplying rather than adding/subtracting. (multiplying by 3)*
- *Pattern C- This pattern is decreasing and not increasing. (decreasing by 3)*
- *Pattern D- This pattern does not start with a 5.*

2. Find a classmate who selected a different pattern that doesn't belong. Write their explanation in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary of their explanation stating that your summary is correct.

My summary of _____'s description	Initials



10.10.2 Guided Practice: Exploring Tables to Determine Linear Functions

- Complete the description observed in the table of values for each function.

Function		Description												
A	<table><tr><th>x</th><th>y</th></tr><tr><td>1</td><td>19</td></tr><tr><td>2</td><td>13</td></tr><tr><td>3</td><td>7</td></tr><tr><td>4</td><td>1</td></tr><tr><td>5</td><td>-5</td></tr></table>	x	y	1	19	2	13	3	7	4	1	5	-5	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
	x	y												
	1	19												
	2	13												
	3	7												
4	1													
5	-5													
B	<table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>4</td></tr><tr><td>-2</td><td>12</td></tr><tr><td>-1</td><td>36</td></tr><tr><td>0</td><td>108</td></tr><tr><td>1</td><td>324</td></tr></table>	x	y	-3	4	-2	12	-1	36	0	108	1	324	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
	x	y												
	-3	4												
	-2	12												
	-1	36												
0	108													
1	324													
C	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>21</td></tr><tr><td>-1</td><td>12</td></tr><tr><td>0</td><td>7</td></tr><tr><td>1</td><td>6</td></tr><tr><td>2</td><td>9</td></tr></table>	x	y	-2	21	-1	12	0	7	1	6	2	9	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
	x	y												
	-2	21												
	-1	12												
	0	7												
1	6													
2	9													
D	<table><tr><th>x</th><th>y</th></tr><tr><td>-4</td><td>-8</td></tr><tr><td>-2</td><td>-1</td></tr><tr><td>0</td><td>6</td></tr><tr><td>2</td><td>13</td></tr><tr><td>4</td><td>20</td></tr></table>	x	y	-4	-8	-2	-1	0	6	2	13	4	20	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
	x	y												
	-4	-8												
	-2	-1												
	0	6												
2	13													
4	20													

10.10.2 GUIDED PRACTICE: EXPLORING TABLES TO DETERMINE LINEAR FUNCTIONS

TEACHER MOVES

Students may need to refer back to Unit 7, where they determined the slope of a line of a linear relationship given a table of values. Emphasize that the notation $\Delta y/\Delta x$ represents the rate of change.

DIFFERENTIATE INSTRUCTION

For a different entry point, have students first write the change of each x-value to the left of the table and the change of each y-value to the right of the table. Students can then circle where the change was not constant.

TEACHER MOVES

Think-Pair-Share

Initially, have students attempt the questions individually. Next, pair students up and ask them to discuss their strategies, reasoning, and answers. Tell students to resolve any differences in their answers. This routine helps students build their confidence and understanding.

ENRICHMENT

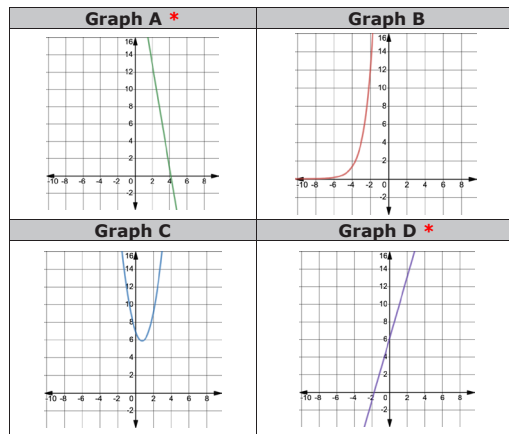
How does the difference between the x-values in Table D compare to those that are in the other tables? Does this affect the answer?

10.10.2 GUIDED PRACTICE: EXPLORING TABLES TO DETERMINE LINEAR FUNCTIONS (CONTINUED)

QUESTIONING

Students can use previous lessons to identify what the equations may look like for Graphs B and C. They may not recognize the exact equation but look for Graph B having a variable as an exponent and Graph C having an x exponent of 2.

2. For each table that has a constant rate of change, complete the statement.
 - a. In table A as the x -values increase by 1, the y -values decrease by 6.
 - b. In table D as the x -values increase by 2, the y -values increased by 7 (or as the x -values increase by 1, the y -values increase by 3.5).
3. The graphs for each of the functions represented by tables are shown. Put an asterisk (*) next the graphs that are linear functions.





10.10.3 Guided Instruction: Identifying Linear Functions from Tables

- 1. Notice that for the linear functions, as the x -values in the tables increase by the same amount, the y -values either increase or decrease by the same amount.
- 2. In each example of a linear function, the rate of change, $\frac{\Delta y}{\Delta x}$, is constant.



A **linear function** is a function that has a constant rate of change.

- 3. Determine which tables of values show a rate of change, $\frac{\Delta y}{\Delta x}$, that is constant.

Table E		Table F		Table G	
x	y	x	y	x	y
0	1	-3	3	2	-4
1	4	0	1	4	0
2	7	3	-1	6	4
3	10	9	-5	8	8

All of the tables of values have a rate of change that is constant.

- 4. The graphs for each of the functions E, F, and G are shown with their corresponding table of values. If the function represents a linear function, circle the function's letter.

Function	Table	Graph										
E	<table><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>7</td></tr><tr><td>3</td><td>10</td></tr></tbody></table>	x	y	0	1	1	4	2	7	3	10	
x	y											
0	1											
1	4											
2	7											
3	10											
F	<table><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-3</td><td>3</td></tr><tr><td>0</td><td>1</td></tr><tr><td>3</td><td>-1</td></tr><tr><td>9</td><td>-5</td></tr></tbody></table>	x	y	-3	3	0	1	3	-1	9	-5	
x	y											
-3	3											
0	1											
3	-1											
9	-5											
G	<table><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>2</td><td>-4</td></tr><tr><td>4</td><td>0</td></tr><tr><td>6</td><td>4</td></tr><tr><td>8</td><td>8</td></tr></tbody></table>	x	y	2	-4	4	0	6	4	8	8	
x	y											
2	-4											
4	0											
6	4											
8	8											

- 5. In Function E, describe what is different in the graph that was not shown in the table.
The graph is a v-shape (absolute value), but the table only showed one side of the graph.
- 6. In Function G, describe what is different in the graph that was not shown in the table.
The graph is broken into pieces and the table only shows one piece of the graph.

A table of values only gives some of the points on the graph of a function. This is not always enough information to conclude that a function is linear. Therefore, from a table of values, it can only be concluded that a function *could* represent a linear function.

10.10.3 GUIDED INSTRUCTION: IDENTIFYING LINEAR FUNCTIONS FROM TABLES

DIFFERENTIATE INSTRUCTION

Consider having students create a reference card with an explanation of how determine if a table of values, equation, or graph is a linear function. Encourage students to include at least one example of each process. Allow students to use their reference cards during the lesson.

TEACHER MOVES

Use Table F to show how the rate of change is equivalent between each point, even when the increments are not equal.

ENRICHMENT

- How does Graph F illustrate the definition “a linear function is a function that has a constant rate of change”?
- How are Graphs E and G counterexamples of “a linear function is a function that has a constant rate of change”?

DIFFERENTIATE INSTRUCTION

Consider pairing students up for questions 5 and 6. Students can share what they notice as being different in the graph that was not in the table. Giving students a chance to first verbalize their observations gives some students an entry point to the writing.

10.10.4 YOUR TURN!

ENRICHMENT

Why do the directions read “could be linear functions”? (A table of values is just a subset of all of the points of a function. Refer back to Graph G to show students that a table of values for this graph included only the values from a linear piece of the three pieces.)

QUESTIONING

For any tables that cannot represent a linear function, change the values so that the table could represent a linear function.



10.10.4 Your Turn!

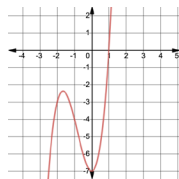
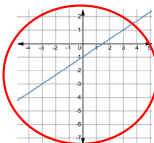
- Determine which relations could be linear functions. Show your work.

Relation	Description										
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-7</td><td>4</td></tr> <tr> <td>-3</td><td>3</td></tr> <tr> <td>1</td><td>2</td></tr> <tr> <td>7</td><td>1</td></tr> </tbody> </table>	x	y	-7	4	-3	3	1	2	7	1	The relation ☐ could be a linear function because ... <i>the rate of change between (1, 2) and (7, 1) is not equal to -1/4, as it is between the other points.</i> ☒ cannot be
x	y										
-7	4										
-3	3										
1	2										
7	1										
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>12</td><td>-2</td></tr> <tr> <td>8</td><td>1</td></tr> <tr> <td>4</td><td>4</td></tr> <tr> <td>0</td><td>7</td></tr> </tbody> </table>	x	y	12	-2	8	1	4	4	0	7	The relation ☐ could be a linear function because ... <i>the rate of change is constant ($-\frac{3}{4}$).</i> ☒ cannot be
x	y										
12	-2										
8	1										
4	4										
0	7										

- Six different relations are shown. Circle the relations that represent linear functions.

$y = 2.5x$

$y = \frac{8}{x} + 1.2$



$y = -4x^3 + 7$

$4.2x = 0.1 - 3y$



10.10.5 Wrap-Up: Ticket Out the Door

1. Complete the table to describe each relation.

Relation	Description										
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>0</td><td>-4</td></tr><tr><td>2</td><td>-8</td></tr><tr><td>4</td><td>-16</td></tr></table>	x	y	-2	-2	0	-4	2	-8	4	-16	The relation ☐ could be ☐ cannot be a linear function because ... <i>the rate of change is constant</i>
x	y										
-2	-2										
0	-4										
2	-8										
4	-16										
<table><tr><th>x</th><th>y</th></tr><tr><td>-5</td><td>6</td></tr><tr><td>-2</td><td>2</td></tr><tr><td>1</td><td>-2</td></tr><tr><td>4</td><td>-6</td></tr></table>	x	y	-5	6	-2	2	1	-2	4	-6	The relation ☐ could be ☐ cannot be a linear function because ... <i>the rate of change is constant ($-\frac{4}{3}$).</i>
x	y										
-5	6										
-2	2										
1	-2										
4	-6										
$y = 7.2$	The relation ☐ is ☐ cannot be a linear function because ... <i>the rate of change is constant (0).</i>										
	The relation ☐ is ☐ cannot be a linear function because ... <i>the graph is not of a function.</i>										

10.10.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

The Wrap-Up asks students to write a justification for why the table of values, graph, or equation is a linear function. Consider providing vocabulary terms that students must use in their responses as scaffolding. For example, students must use the phrase “rate of change” in their response to the first relation.